



Vavuniya Campus of the University of Jaffna
Fourth Examination in Information and Communication Technology - 2016
First Semester - June/July 2017
ICT4171 Practical for Bioinformatics and Computational Biology
Answer All Questions

Time : Two hours

Instructions

- Complete each of the following tasks using Python programming language or any language of your choice.
- Documents related to the questions are saved in the folder named *ICT4171_Final* and placed in your Desktop Folder.
- Create a new folder named with your index number (E.g.: *IT8005*) in the Desktop Folder and save all your files in the new folder.

1. Microarray analysis is a method that uses microchips containing anchored arrays of short DNA elements (known as probes) for the large-scale interrogation of gene expression. You are required to draw a heatmap from data given to you as an analysis of *Acute Lymphocytic Leukemia (ALL)* microarray dataset which consists of differentially expressed genes between two types of leukemia.

- (a) Store the result for each of the patients by loading the molecular biology results (in second column) into a Python dictionary, using the patient ID as the key (in first column) from the file *molecular_biology.tsv*.

[To be continued...]

- (b) Store the expression levels for this gene ALL by loading expression data from file *AllExpression.tsv*.
 - (c) Print the number of patients and the number of genes in the given data set.
 - (d) Plot the heatmap using microarray data given.
2. A restriction map has been digested using PstI enzyme into several fragments. Determine the possible restriction maps that can be constructed using the data given below
- $$\Delta X = \{1, 1, 2, 2, 2, 3, 3, 4, 4, 5, 5, 5, 6, 7, 7, 7, 8, 9, 10, 11, 12\}$$